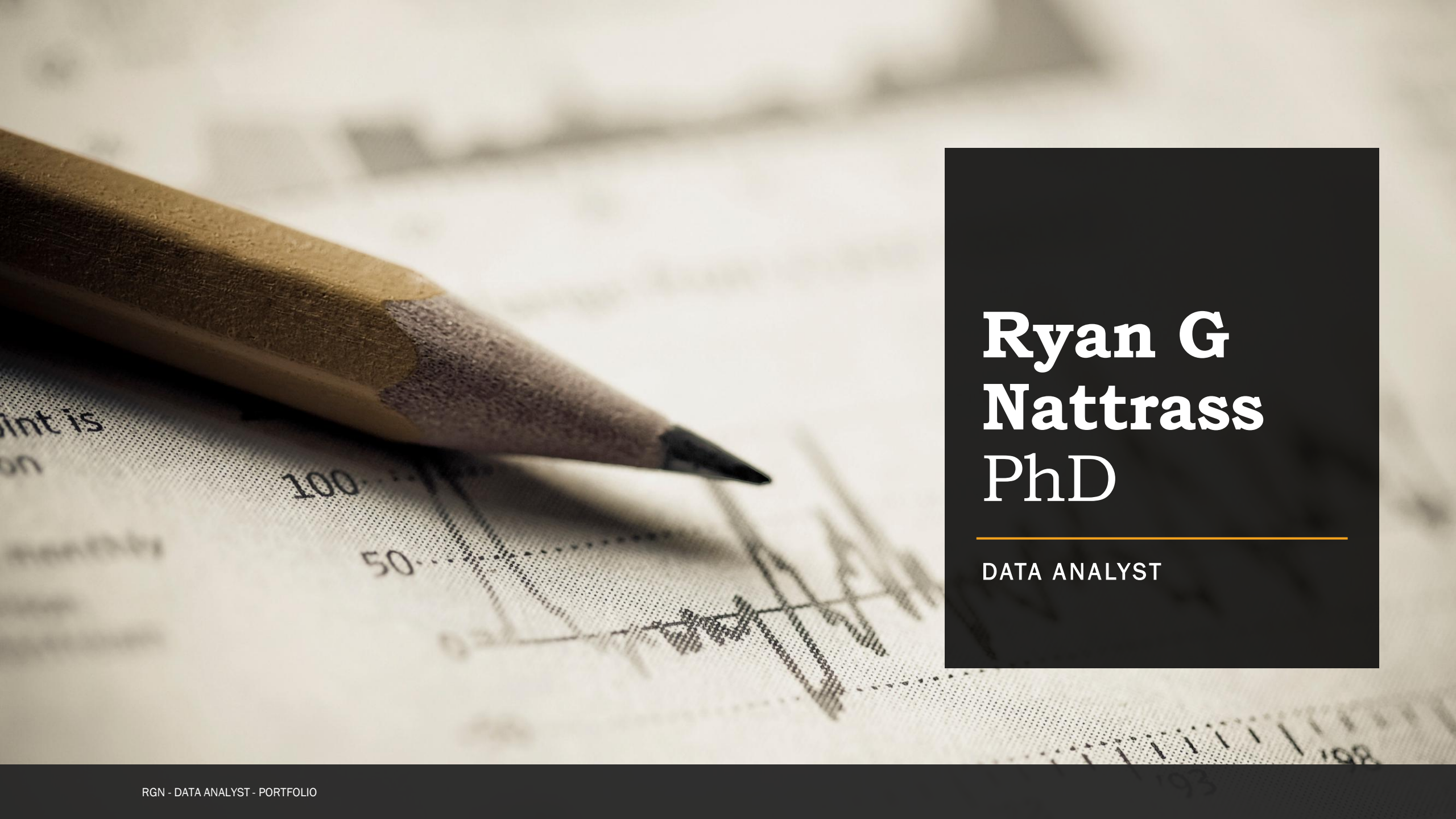




Ryan G Nattrass PhD

DATA ANALYST

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Instacart User Profiling and Analysis



Background: An online grocery store ‘Instacart’ would like understanding customer demographics and behavior in order to improve targeted advertising, and determine how different customers profiles impact customer behavior

Objective:

- Understand Instacart’s busiest periods for advertising purposes
- Determine and analyse customer loyalty levels to provide insights into marketing decisions
- Design customer profiles and analyse buying trends

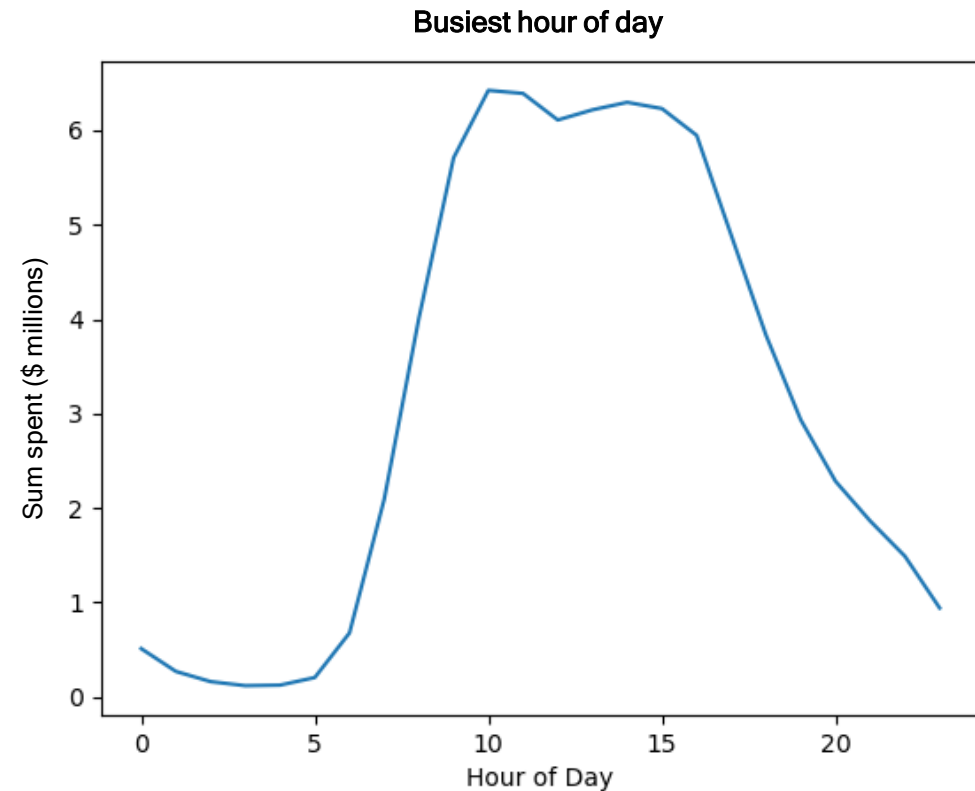
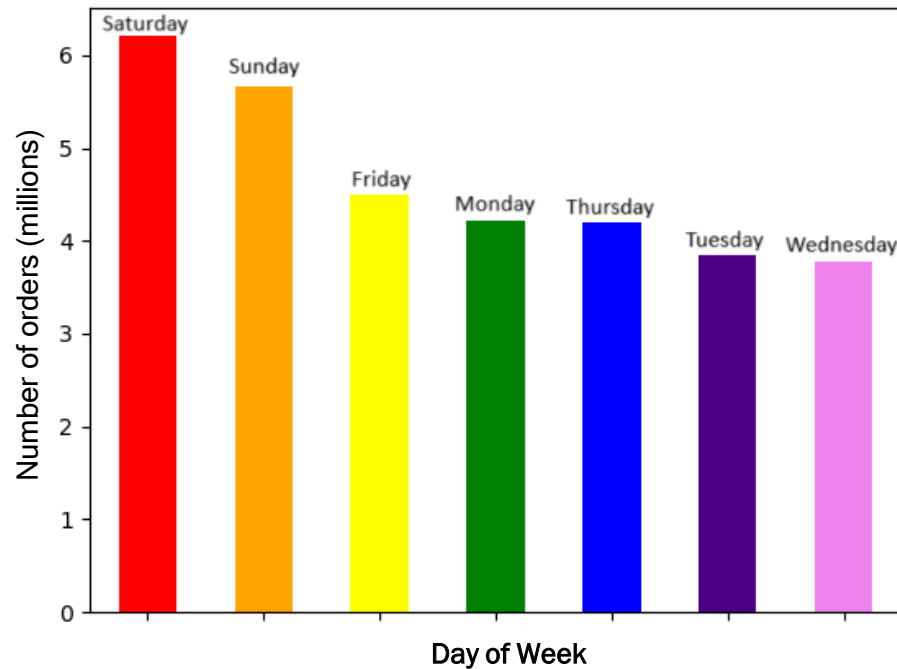
Dataset: “The Instacart Online Grocery Shopping Dataset 2017”, Accessed from www.kaggle.com/inmurs/instant-dataset via Kaggle

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Instacart

Peak ordering times

The most orders for Instacart come during the weekend

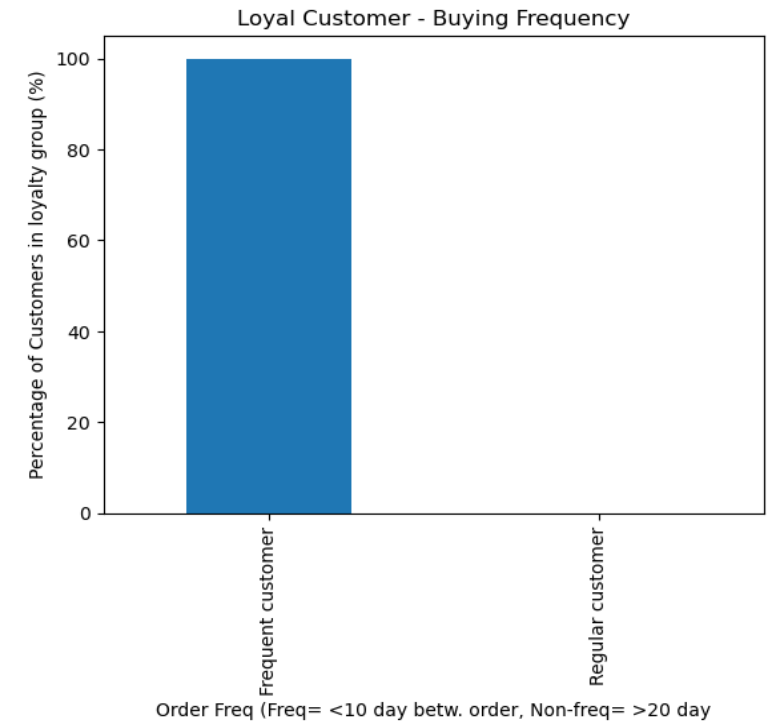
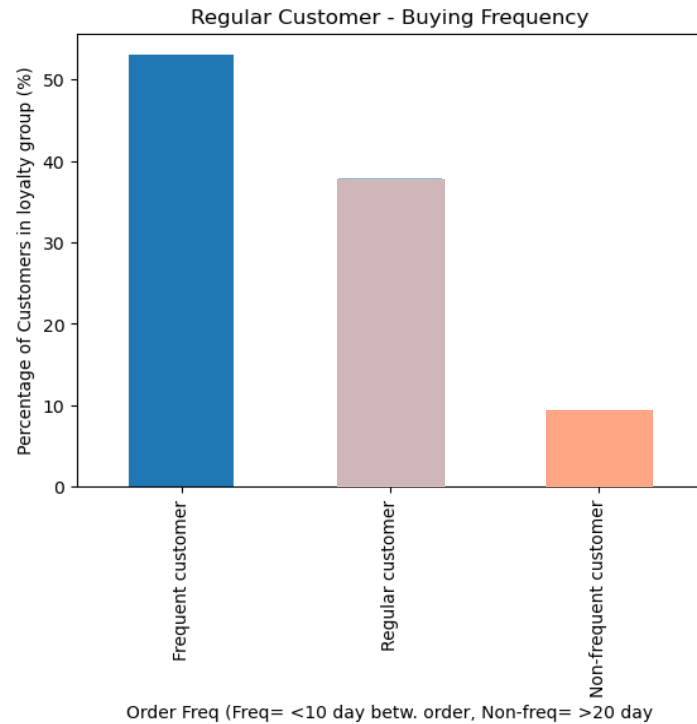
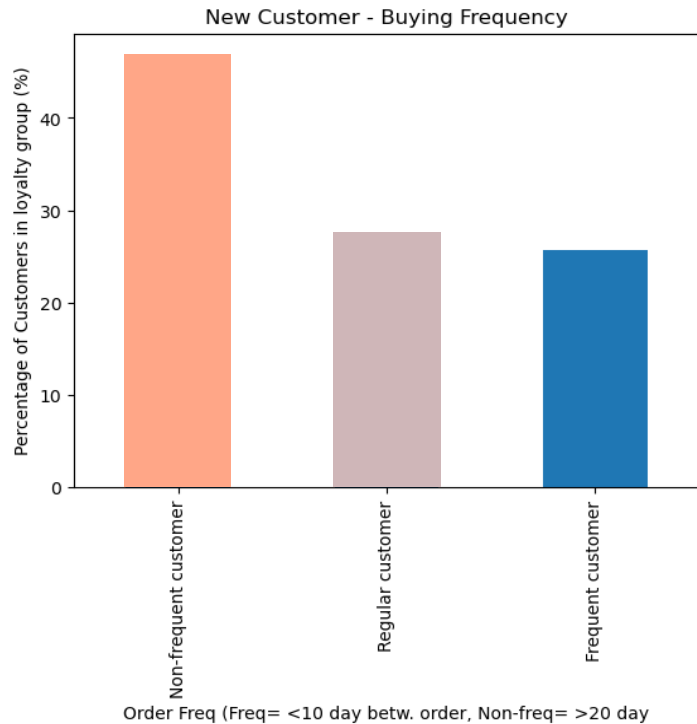


Peak-daily-orders are between 10am-4pm

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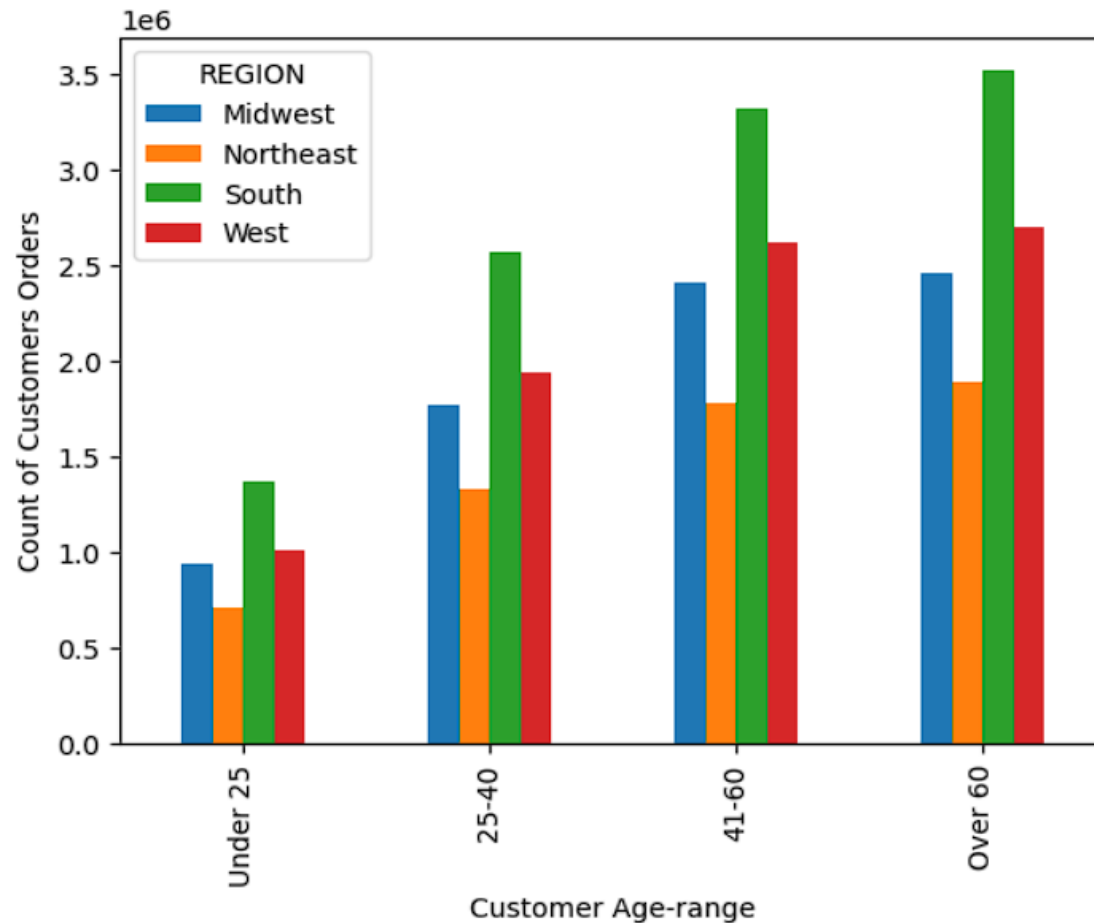
Customer-Loyalty and impact on orders

Less Loyal, Less-frequent orders ← → More Loyal, More-frequent orders



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Instacart Biggest customer base

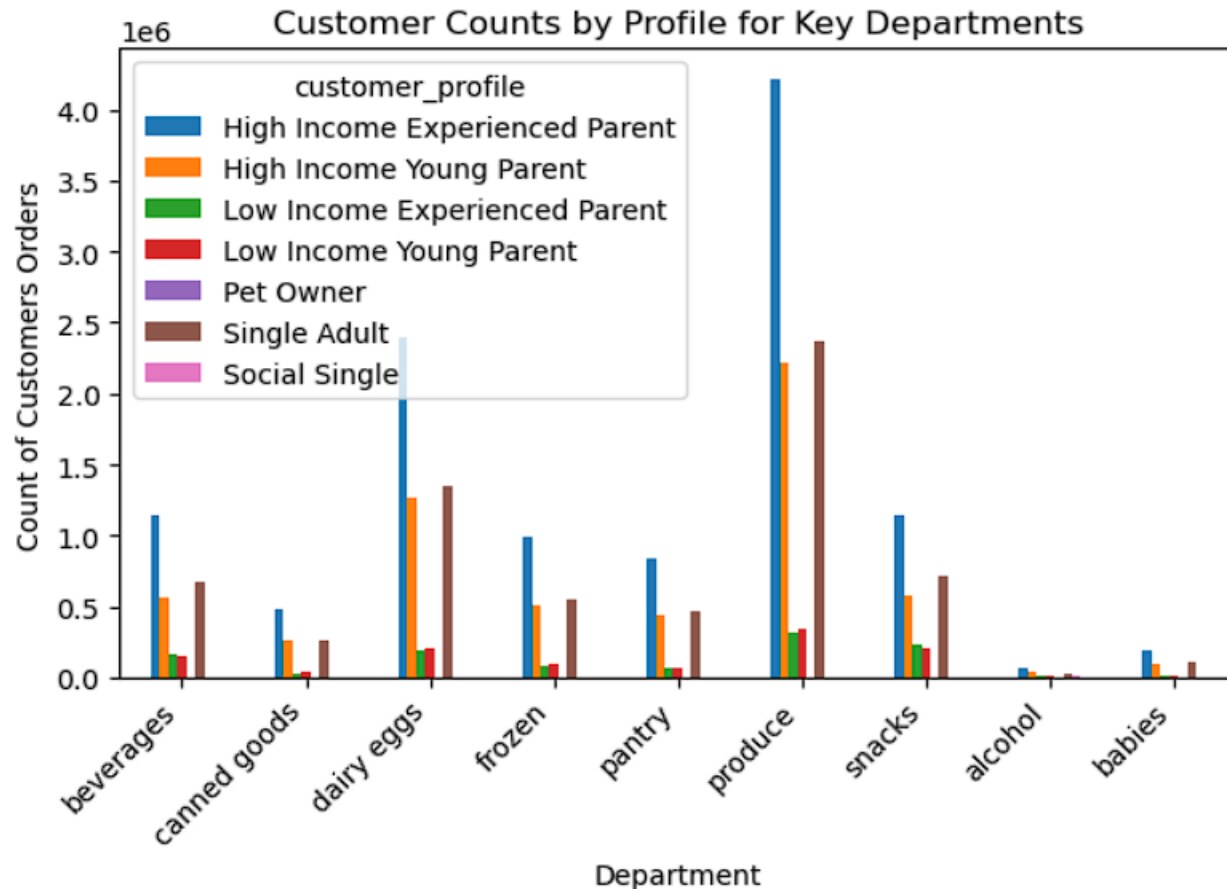


Instacart's biggest customer base:

- Is over the age of 40
- Resides in the south of the USA

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Customer Profiling & Most-popular department



The most customers that use Instacart are:

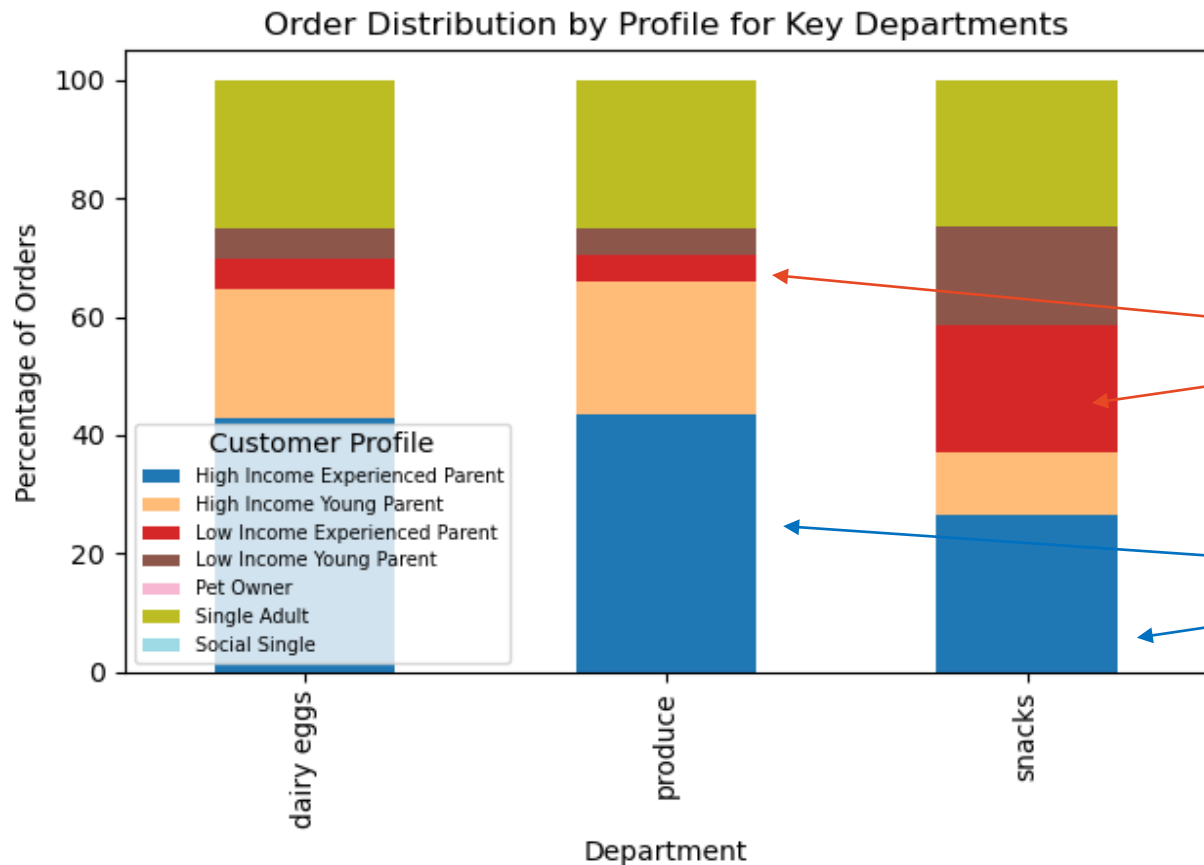
1. High-income parents >40
2. Single Adults
3. High-income parents < 40

The most orders come from the departments:

1. Produce
2. Dairy/ eggs
3. Snacks
4. Beverages

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Customer profiles and ordering habits



While produce is the most ordered-from department for most customer-profiles, **low-income families** are much more likely to order from the snacks department than other customer-profiles

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Instacart Data-driven recommendations



- To maximise on the time most customers use Instacart the least, scheduled targeted-ads should run in the evenings of weekdays or before 10am on weekends.
- Data shows that the more orders a customer has completed with Instacart, the more likely they are to order again, therefore it may be beneficial to incentivize customers to make more orders
- With regards to targeted ads, the largest customer base Instacart has are parents older than 40 and have more than 50,000 annual income
- Produce is most-often ordered by everyone except low-income families who prefer to use Instacart to order snacks